

UniPi : Learning Universal Policies via Text-Guided Video Generation

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1. Introduction

Background:

Large pretrained models in NLP & CV enable **zero-shot generalization**.

Next step: build **generalist agents** that solve decision-making tasks across diverse environments.

Problem:

Environments have **heterogeneous state-action spaces** (e.g., MuJoCo continuous control vs. Atari discrete actions).

Heterogeneity impedes **knowledge sharing, learning, and generalization**; **reward design** is also difficult.

Encoding everything with **universal tokens** in sequence models may fail to preserve and transfer rich V&L prior knowledge.

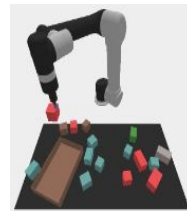
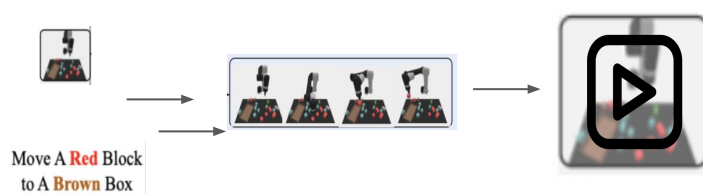
Key Idea: A Universal Interface

Video (image sequences) as a common medium for observations & actions.

Text as a universal medium for task specification.

UniPi Pipeline

- 1) Input current frame + goal text → **text-conditioned video diffusion** generates a **future trajectory (video)**.
- 2) Apply **inverse dynamics** to the generated frames to **regress action sequences**.
- 3) Use regressed actions to build and **execute the policy**.



1. Introduction

Advantages

Combinatorial Generalization

Leverage language compositionality to synthesize **unseen object/relationship configurations**.

Multi-task Learning

Casting **action prediction as video prediction** enables joint learning over many tasks and **test-time zero-shot** generalization

Action planning

Hierarchical: Generate coarse, temporally sparse frames then **refine** into a detailed plan.

Steerable: Inject constraints/preferences via **test-time sampling**.

Interpretable: Plans are **videos**, making verification and diagnosis intuitive.

Internet-Scale Knowledge Transfer

Pretrain on large web text–video corpora to recover **demonstrations**, enabling realistic robot motion synthesis from language

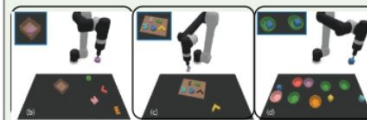
Main Contribution

Formulate **text-conditioned video generation** as a **universal planning strategy** to synthesize diverse behaviors in one framework. Demonstrate **generalization advantages** over direct action-prediction RL baselines across domains (see Fig. 1 & §4).

Combinatorial Language Long-Horizon Planning

Wipe A **Brown** Box
Before Pick up A **Red**
Block. Then Put Red
Block on Brown Box

Multi-Task Generalization



Internet-Scale Knowledge Transfer

Wipe the
Counter with
Robot Arm and
Red Cloth



2. Problem Statement

Limitation of MDP $\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, T, R, \gamma \rangle$

State Representation Limitation $\forall E_1, E_2 : \mathcal{S}_{E_1} \neq \mathcal{S}_{E_2}$

No universal state interface across environments

Reward Function Limitation $R : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$

Rewards must be real-valued and environment-specific, making design and transfer difficult.

Reward Function Limitation $T(s'|s, a) : \mathcal{S} \times \mathcal{A} \rightarrow \Delta(\mathcal{S})$

Transition dynamics are environment- and agent-dependent, hindering knowledge transfer

Unified Predictive Decision Process (UPDP) $G = \langle X, C, H, \rho \rangle$

X : Observation space of images $H \in \mathbb{N}$: Finite horizon length

C : Space of textual task descriptions $\rho(\cdot|x_0, c) : X \times C \rightarrow \Delta(X^H)$: Conditional video generator producing H - step trajectories

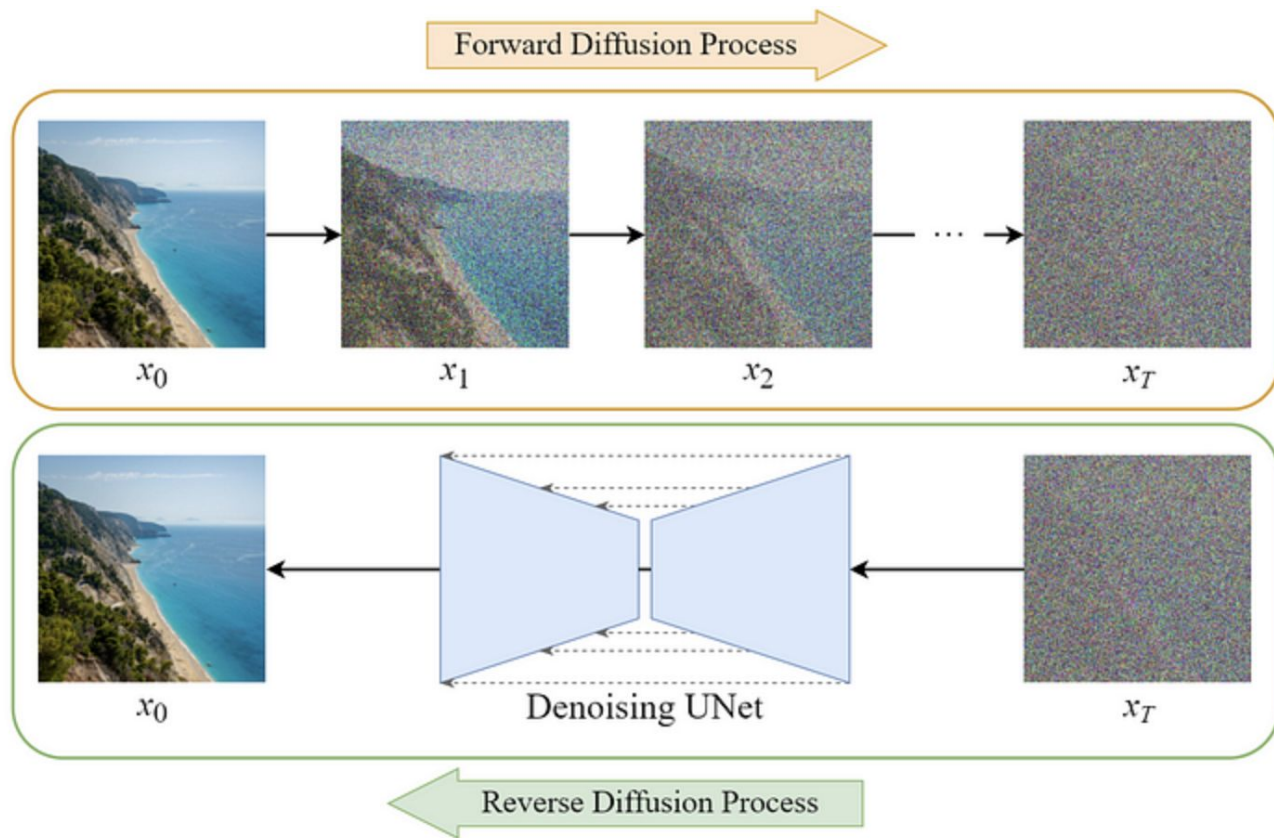
Trajectory-Task Conditioned Policy $\pi(\cdot|\{x_h\}_{h=0}^H, c) : X^{H+1} \times C \rightarrow \Delta(\mathcal{A}^H)$

Input : Trajectory $\{x_h\}_{h=0}^H$ and task C

Output : Distribution over H -step action sequences $\mathcal{A}^H$ such that aligns actions with the generated trajectory from ρ

2. Problem Formulation

Diffusion model



2. Problem Formulation

Unconditional Model

Continuous-time diffusion model

Forward Process $q_k(\tau_k|\tau) = \mathcal{N}(\cdot; \alpha_k\tau, \sigma_k^2 I)$

τ : the original data (in this case, a video sequence). $\tau = [x_1, \dots, x_H] \in X^H$

τ_k : the data with noise added at step k .

q_k : the probability distribution at step k of the forward (noising) process.

$\mathcal{N}$: Gaussian distribution (mean = $\alpha_k\tau$, variance = σ_k^2).

Role: the formula that gradually transforms a clean image τ into noise.

Reverse Process $p(\tau)$

denoising-model $s(\tau_k, k)$

Conditioning

conditional denoiser $s(\tau_k, k|c, x_0)$

Classifier-free guidance

$$\hat{s} = (1 + \omega) s(\tau_k, k|c, x_0) - \omega s(\tau_k, k)$$

$$\hat{s} = \underbrace{s(\tau_k, k|c, x_0)}_{\text{conditional output}} + \omega \left(\underbrace{s(\tau_k, k|c, x_0)}_{\text{conditional output}} - \underbrace{s(\tau_k, k)}_{\text{unconditional output}} \right)$$

$s(\tau_k, k | c, x_0)$: **conditional output**
(incorporating the text instruction and the initial frame)

$s(\tau_k, k)$ **unconditional output**
(reconstruction without any conditioning)

ω : the strength parameter that controls
how strongly the conditioning is emphasized

3. Unified Mathematical Specification of the Video U-Net

Input and indexing :

The input video is $\mathbf{x} \in \mathbb{R}^{B \times T \times H \times W \times C}$ with indices (b, t, h, w, c) the model has L scales.

At scale l , the spatial size is $(H_l, W_l) = (H/2^l, W/2^l)$, and the channel width is $C_l = C_{\text{base}} \cdot m_l$, $C_{\text{base}} = 512$, $m_l \in \{1, 2, 4\}$

First-frame conditioning :

The first frame of each video is replicated across all time indices: $\tilde{\mathbf{x}}_{b,t,h,w,c} = \mathbf{x}_{b,1,h,w,c}$, $\forall b \in [B], t \in \{1, \dots, T\}$

Channel-wise concatenation is applied to the noisy input $\mathbf{y}_{b,t,h,w,:} = [\mathbf{x}_{b,t,h,w,:}^{\text{noisy}} \parallel \tilde{\mathbf{x}}_{b,t,h,w,:}]$, so that $\mathbf{y} \in \mathbb{R}^{B \times T \times H \times W \times 2C}$

An input projection conv layer maps $\mathbf{y}$ into C_0 channels : $\mathbf{z}^{(0)} = \text{Conv}^{(0)}(\mathbf{y})$

Residual block :

At scale l , for residual block $k \in \{1, 2, 3\}$, with normalization $\mathcal{N}$ and nonlinearity σ :

$$\mathbf{r}_{b,t,h,w,:}^{(l,k)} = \text{Conv}^{(l,k,2)} \left(\sigma \left(\mathcal{N} \left(\text{Conv}^{(l,k,1)} \left(\sigma \left(\mathcal{N} \left(\mathbf{h}_{b,t,h,w,:}^{(l,k-1)} \right) \right) \right) \right) \right) \right), \mathbf{h}_{b,t,h,w,:}^{(l,k)} = \mathbf{h}_{b,t,h,w,:}^{(l,k-1)} + \mathbf{r}_{b,t,h,w,:}^{(l,k)}$$

Temporal convolution :

After each residual block, local temporal mixing is applied: $\tilde{\mathbf{h}}_{b,t,h,w,:}^{(l,k)} = \sum_{\Delta=-\kappa}^{\kappa} U_{\Delta}^{(l,k)} \mathbf{h}_{b,t+\Delta,h,w,:}^{(l,k)}$, then update $\mathbf{h}_{b,t,h,w,:}^{(l,k)} \leftarrow \tilde{\mathbf{h}}_{b,t,h,w,:}^{(l,k)}$

Attention at selected resolutions :

For scales with $(H_l, W_l) \in \{(6, 6), (12, 12), (24, 24)\}$, multi-head self-attention is applied: $\text{Attn}(\mathbf{h}) = \text{Proj} \left(\text{Concat}_{i=1}^{n_h} \left[\text{softmax} \left(\frac{Q_i K_i^{\top}}{\sqrt{d_h}} \right) V_i \right] \right)$
with $d_h = 64$, $n_h = C_l/d_h$, $Q_i = \mathbf{h}W_i^Q$, $K_i = \mathbf{h}W_i^K$, $V_i = \mathbf{h}W_i^V$

3. Unified Mathematical Specification of the Video U-Net

Encoder (contracting path) :

At each scale $l = 0, \dots, L - 1$, apply 3 residual blocks (with temporal conv and optional attention). The output $\mathbf{s}^{(l)} = \mathbf{h}^{(l,3)}$ is stored as a skip connection, and downsampling by stride-2 conv gives the input to the next scale: $\mathbf{h}^{(l+1,0)} = \text{Down}^{(l)}(\mathbf{h}^{(l,3)})$

Decoder (expanding path) :

At the top scale, $\mathbf{g}^{(L,0)} = \mathbf{h}^{(L,3)}$, for $l = L - 1, \dots, 0$, upsampling is performed: $\hat{\mathbf{g}}^{(l,0)} = \text{Up}^{(l)}(\mathbf{g}^{(l+1,3)})$

Then skip features are concatenated: $\tilde{\mathbf{g}}^{(l,0)} = [\hat{\mathbf{g}}^{(l,0)} \parallel \mathbf{s}^{(l)}]$. Three residual blocks (with temporal conv and optional attention) are then applied to $\tilde{\mathbf{g}}^{(l,0)}$ to produce $\tilde{\mathbf{g}}^{(l,3)}$, which is passed to the next scale.

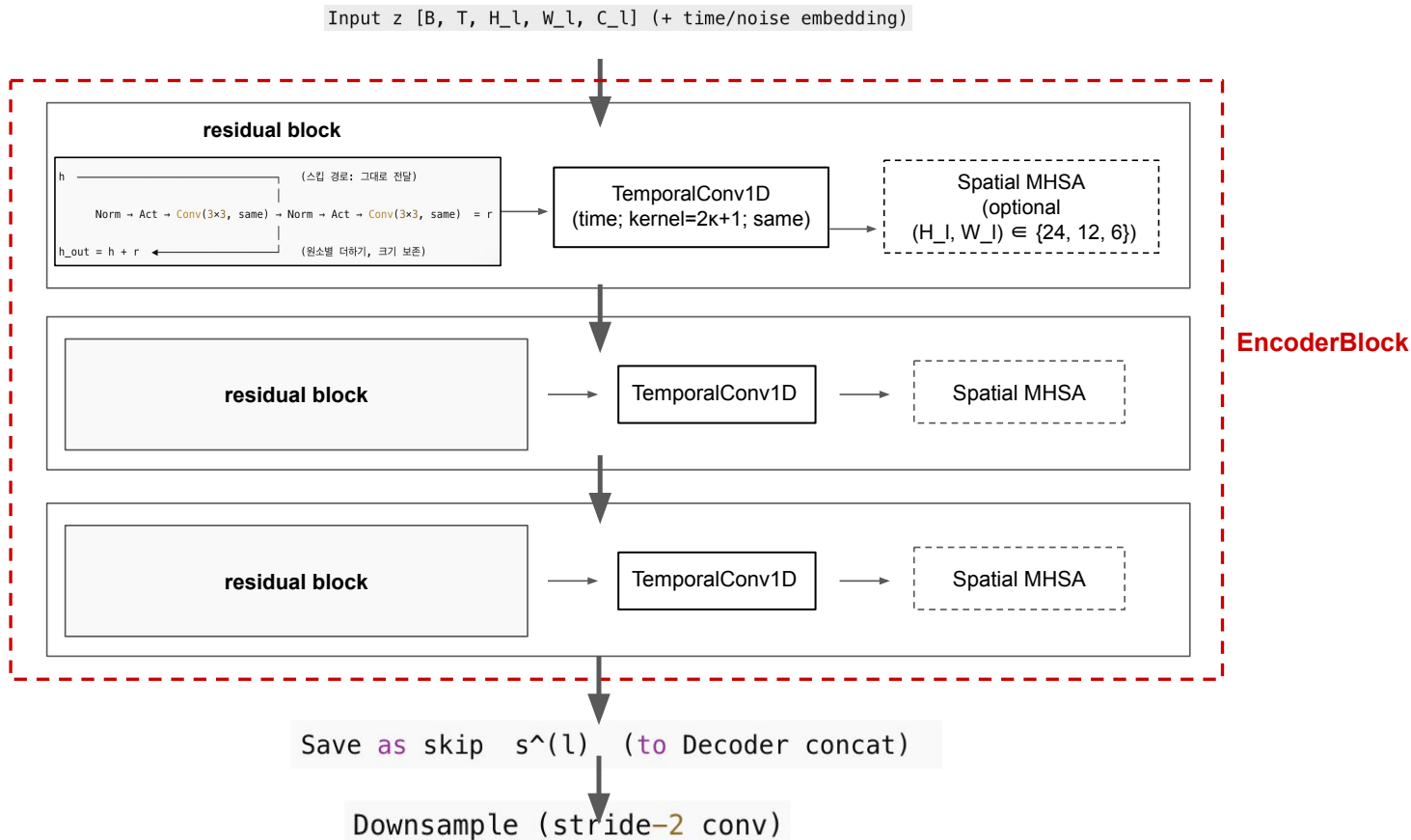
Output head :

Finally, a 1×1 conv maps $C_0 \rightarrow C$ channels: $\hat{\mathbf{x}}_{b,t,h,w,c} = \sum_{c'} W_{c',c}^{\text{out}} \mathbf{g}_{b,t,h,w,c'}^{(0,3)} + b_c^{\text{out}}$

Summary :

The overall model is: $\hat{\mathbf{x}} = f_{\theta}(\mathbf{x}^{\text{noisy}}, \tilde{\mathbf{x}}, \mathbf{c}) = \text{OutHead} \circ \left(\prod_l \text{DecBlock}_l \right) \circ \text{Bottleneck} \circ \left(\prod_l \text{EncBlock}_l \right) \left(\text{InProj}([\mathbf{x}^{\text{noisy}} \parallel \tilde{\mathbf{x}}]) \right)$

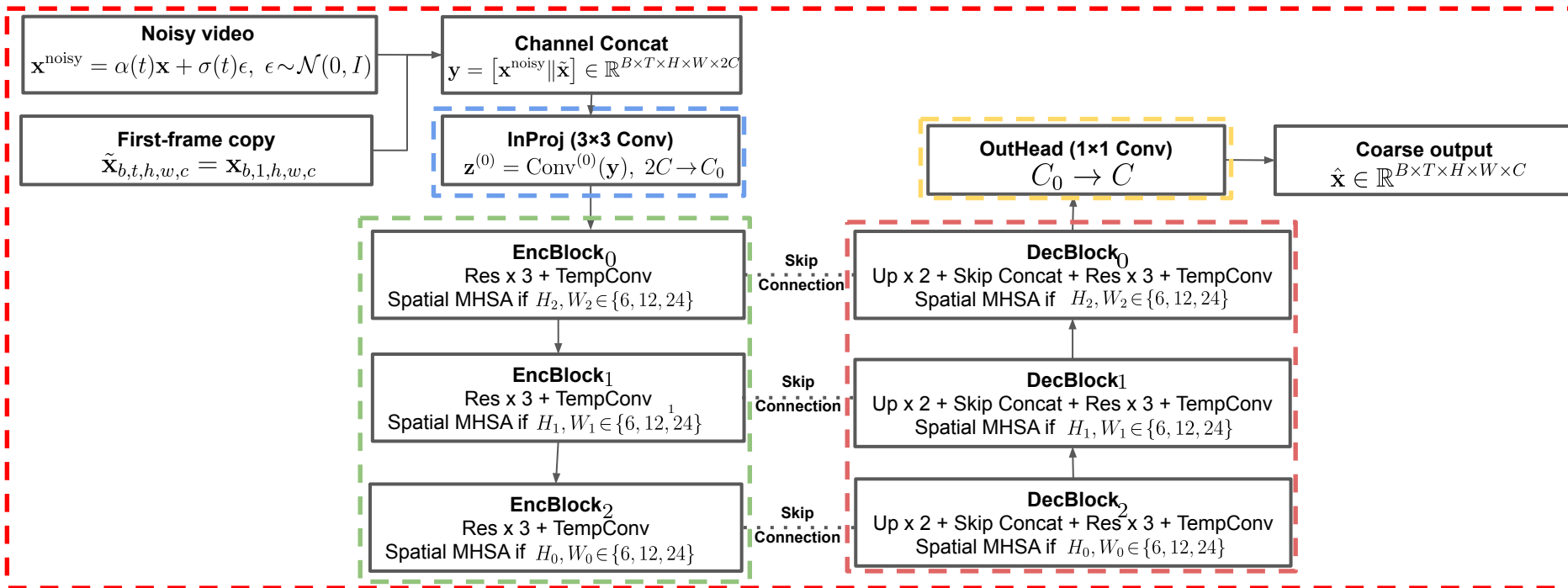
3. Unified Mathematical Specification of the Video U-Net



3. Abstract Structure of Video U-Net

Summary :

The overall model is: $\hat{\mathbf{x}} = f_{\theta}(\mathbf{x}^{\text{noisy}}, \tilde{\mathbf{x}}, \mathbf{c}) = \text{OutHead} \circ \left(\prod_l \text{DecBlock}_l \right) \circ \text{Bottleneck} \circ \left(\prod_l \text{EncBlock}_l \right) \left(\text{InProj}([\mathbf{x}^{\text{noisy}} \parallel \tilde{\mathbf{x}}]) \right)$



3. Detail - Inverse Dynamics Training Details

Inverse dynamics model

: MLP directly predict the 7-dimensional controls of the simulated robot arm from an image observation mean squared error.
inverse dynamics model consists of a 3x3 convolutional layer, 3 layers of 3x3 convolutions with residual connection, a mean-pooling layer across all pixel locations, and an MLP layer of (128, 7) channels to predict the final controls.

$$f_{\theta}(\mathbf{o}) = \text{MLP}\left(\text{MeanPool}\left(\text{ResConv}_3\left(\text{Conv}_{3 \times 3}(\mathbf{o})\right)\right)\right)$$

(1) Convolution layer (3x3)

: For in $\mathbf{h} \in \mathbb{R}^{H \times W \times C}$, $\text{Conv}_{3 \times 3}(\mathbf{h})(i, j) = \sum_{u=-1}^1 \sum_{v=-1}^1 W_{uv} \mathbf{h}(i+u, j+v) + b$, where W_{uv} are learnable kernels and b is a bias term.

(2) Residual block (3 layers)

: stack three convolutional layers with residual connections, for each layer k : $\mathbf{h}_{k+1} = \mathbf{h}_k + \sigma(\text{Conv}_{3 \times 3}(\mathbf{h}_k))$, after three layers $\text{ResConv}_3(\mathbf{h}) = \mathbf{h}_3$

(3) Mean pooling

: average across spatial positions: $\text{MeanPool}(\mathbf{h}) = \frac{1}{HW} \sum_{i=1}^H \sum_{j=1}^W \mathbf{h}(i, j)$

(4) Multi-layer perceptron (MLP)

: Given a pooled feature vector $\mathbf{z} \in \mathbb{R}^d$, we use a two-layer MLP: $\mathbf{h}' = \sigma(W_1 \mathbf{z} + b_1)$, $W_1 \in \mathbb{R}^{128 \times d}$, $\hat{\mathbf{u}} = W_2 \mathbf{h}' + b_2$, $W_2 \in \mathbb{R}^{7 \times 128}$

Training set up

: Adam optimizer, $\eta = 1 \times 10^{-4}$, $\|\nabla_{\theta} \mathcal{L}\|_2 \leq 1$, 2M steps, a linear warm up schedule is applied for the first 1,000 steps

4. Decision Making with Videos

4.1 Universal Video-Based Planner

Conditional Video Synthesis.

- Train the model with the first frame x_0 always included as a conditioning input.
- At inference, enforce that generation must start from the given initial frame x_0 .

$$\rho(\cdot, [x_0, c]) : X \times C \rightarrow \Delta(X^H)$$

* where x_0 = initial frame, c = textual task description

Trajectory Consistency through Tiling. $\tilde{s}(\tau_k, k | c, x_0) = (1 + \omega) s(\tau_k, k | c, x_0) - \omega s(\tau_k, k)$

- Problem: the environment drifts over time.
- Implementation at sampling: concatenate the first frame (x_0) at every denoising step to preserve the environment.

Hierarchical Planning.

Step 1: Coarse Video Generation: $q_k(\tau_k | \tau) = \mathcal{N}(\cdot; \alpha_k \tau, \sigma_k^2 I)$

- Generate a **sparse sequence of frames** (an abstract video plan / high-level abstraction).

Step 2: Temporal Super-Resolution :

- Densify the time axis to produce **fine-grained video** based on the coarse plan.
- **Interpolate** between frames while reinforcing **environmental and action consistency**.

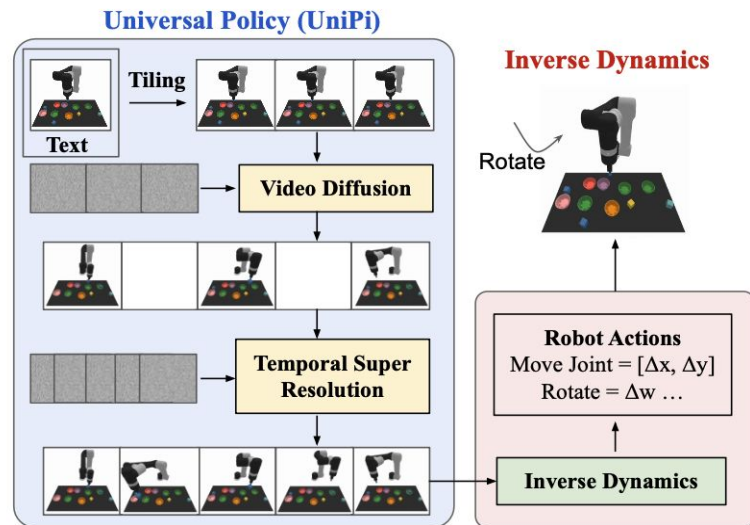
Flexible Behavioral Modulation. prior $h(\tau)$

4.2 Task Specific Action Adaptation

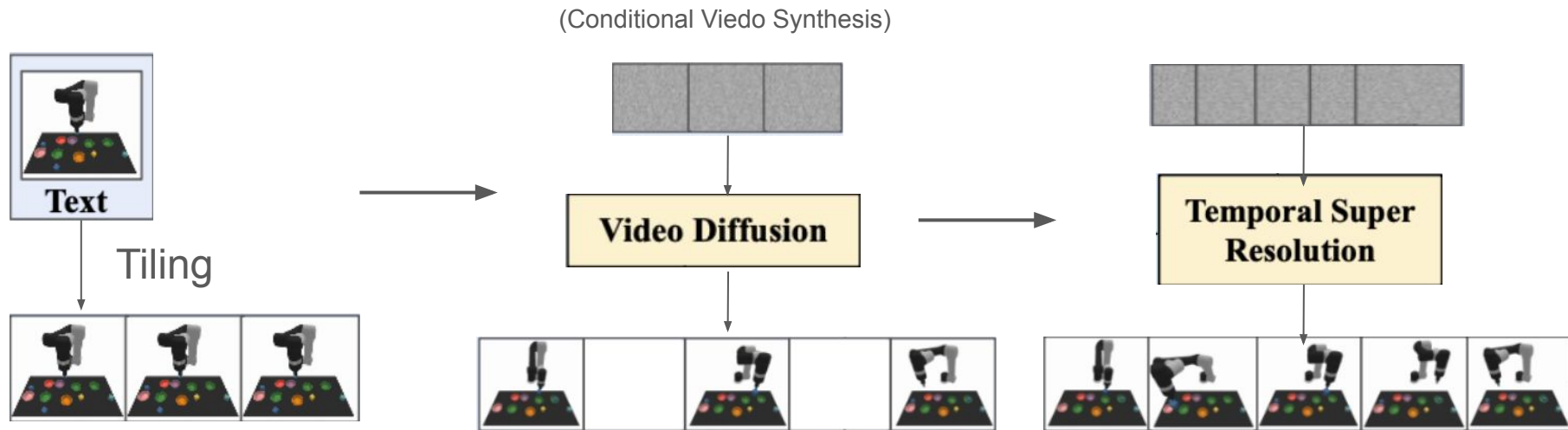
Inverse Dynamics. Infer which action occurred from an input image

Action Execution. Closed-loop : : after each execution step, re-observe the video and recompute a new H-step action sequence—i.e., MPC.

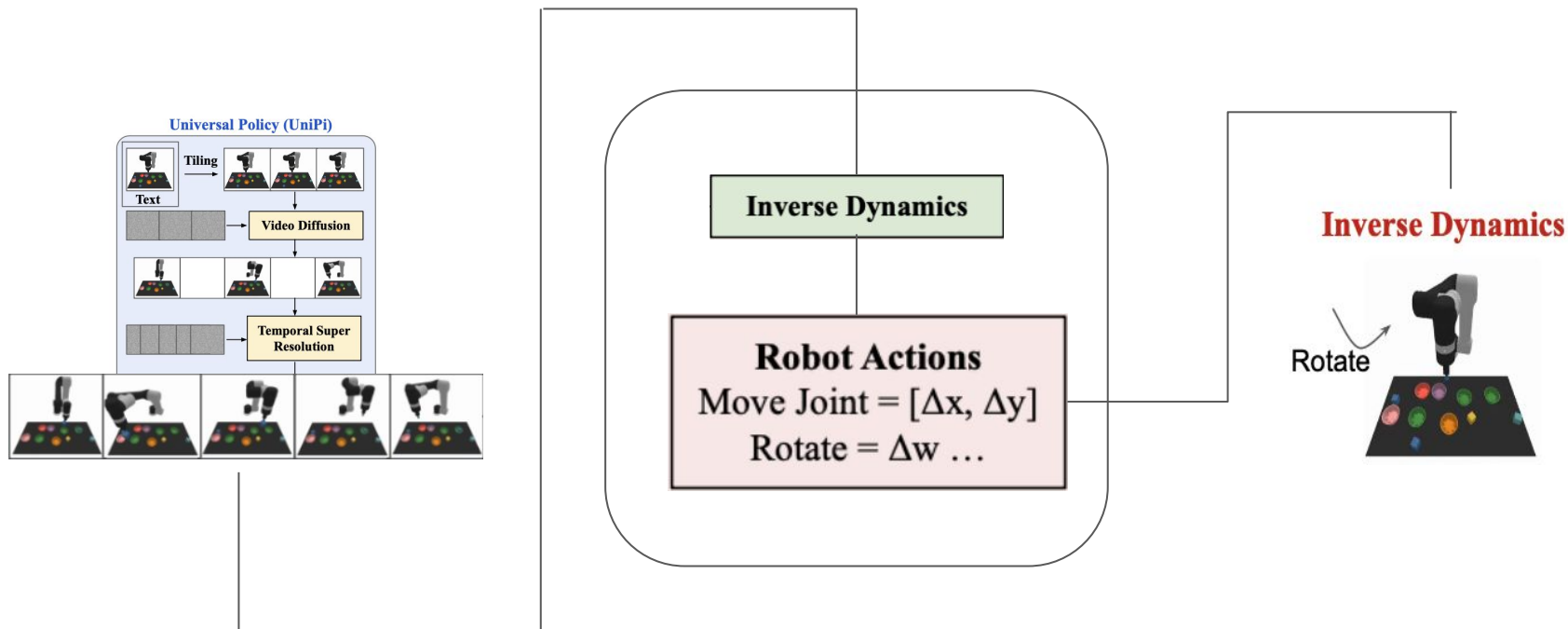
Open-loop (Open-loop only): execute the initially predicted action sequence in order, with no re-planning.



4. Decision Making with Videos



4. Decision Making with Videos



5. Experimental Evaluation

5.1 Combinatorial Policy Synthesis

Goal

Measure UniPi's ability to **combinatorially generalize** to diverse **language-conditioned** tasks.

Environment & Task

Combinatorial robot planning: manipulate blocks to satisfy language relations (e.g., “put a red block right of a cyan block”).

Procedure: pick a **white** block → place into a **color bowl** to paint → place on a **plate** to satisfy the relation.

Randomization: precise locations of blocks/bowls/plates are randomized each episode.

Data: train the video model on **200k** videos of generated language instructions.

Split: **70%** instructions seen at training; **30% unseen** at test time.

Demos constructed with a **scripted agent**.

Baselines

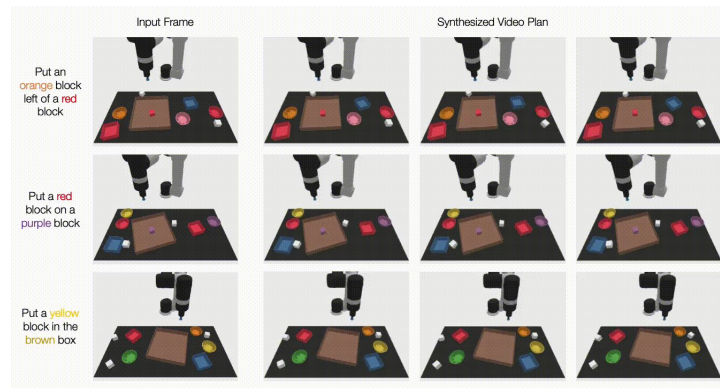
Transformer BC (goal-conditioned):

- **Image + Transformer BC** (obs. images + text → next action)
- **State + Transformer BC** (robot joint state + text → next action)

Trajectory-Transformer-style sequence regression: **Image + TT** (regress a sequence of future actions).

Action-diffusion baseline: image-conditioned diffusion **directly in joint-action space** [21, 27] (vs. UniPi's **video-as-policy** that diffuses future frames).

Note: **Offline RL** baselines not directly applicable (no access to environment reward functions).



5. Experimental Evaluation

5.1 Combinatorial Policy Synthesis

Metrics

- (1) whether the language instruction has been seen during training
- (2) whether the language instruction specifies placing a block in relation to some other block as opposed to direct pick-and-place.

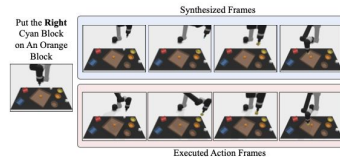


Figure 4: **Action Execution.** Synthesized video plans and executed actions in the simulated environment. The two video plans roughly align with each other.

Ablations

Frame conditioning (first-frame): enforces consistency with the observed initial image.

Frame consistency (tiling): tile the observed frame across timesteps; without it, non-start frames receive a **zeroed image** as context.

Temporal hierarchy (super-resolving over time): yields **more detailed** video plans.

Finding: **All components are crucial**; conditioning/consistency preserve alignment; temporal hierarchy improves detail.

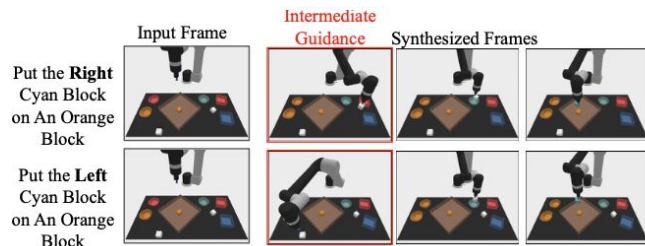
Model	Seen		Novel	
	Place	Relation	Place	Relation
State + Transformer BC [24]	19.4 ± 3.7	8.2 ± 2.0	11.9 ± 4.9	3.7 ± 2.1
Image + Transformer BC [24]	9.4 ± 2.2	11.9 ± 1.8	9.7 ± 4.5	7.3 ± 2.6
Image + TT [25]	17.4 ± 2.9	12.8 ± 1.8	13.2 ± 4.1	9.1 ± 2.5
Diffuser [21]	9.0 ± 1.2	11.2 ± 1.0	12.5 ± 2.4	9.6 ± 1.7
UniPi (Ours)	59.1 ± 2.5	53.2 ± 2.0	60.1 ± 3.9	46.1 ± 3.0

Table1

Frame Condition	Frame Consistency	Temporal Hierarchy	Place	Relation
No	No	No	13.2 ± 3.2	12.4 ± 2.4
Yes	No	No	52.4 ± 2.9	34.7 ± 2.6
Yes	Yes	No	53.2 ± 3.0	39.4 ± 2.8
Yes	Yes	Yes	59.1 ± 2.5	53.2 ± 2.0

Table2

Adaptability



5. Experimental Evaluation

5.2 Multi-Environment Transfer

Goal: Evaluate UniPi’s ability to **learn across many tasks** and **transfer** at test time to **unseen environments**.

Setup Benchmark: language-guided manipulation suite from [28].

Train on **10 tasks** (demonstrations); **test on 3 unseen tasks**.

Data: **200k** language-execution videos generated by a **scripted oracle agent**.

Metric: **Instruction completion accuracy** per prompt (see Appendix A for env details).

Baselines (same as §4.1)

Note on **CLIPort [28]**: not directly comparable—relies on **pick/place primitives** and **pose inference** only; lacks joint-space control and is hard to adapt to this setting.

Findings — Multitask Generalization

UniPi **generalizes** to new tasks/environments, synthesizing valid **video plans** and decisions.

Handles diverse **object categories** and **colors**;

- coherent pick/paint/place sequences from language.

Fig. 5: qualitative video visualizations aligning plans with successful executions.

Model	Place Bowl	Pack Object	Pack Pair
State + Transformer BC	9.8 ± 2.6	21.7 ± 3.5	1.3 ± 0.9
Image + Transformer BC	5.3 ± 1.9	5.7 ± 2.1	7.8 ± 2.6
Image + TT	4.9 ± 2.1	19.8 ± 0.4	2.3 ± 1.6
Diffuser	14.8 ± 2.9	15.9 ± 2.7	10.5 ± 2.4
UniPi (Ours)	51.6 ± 3.6	75.5 ± 3.1	45.7 ± 3.7

Table 3

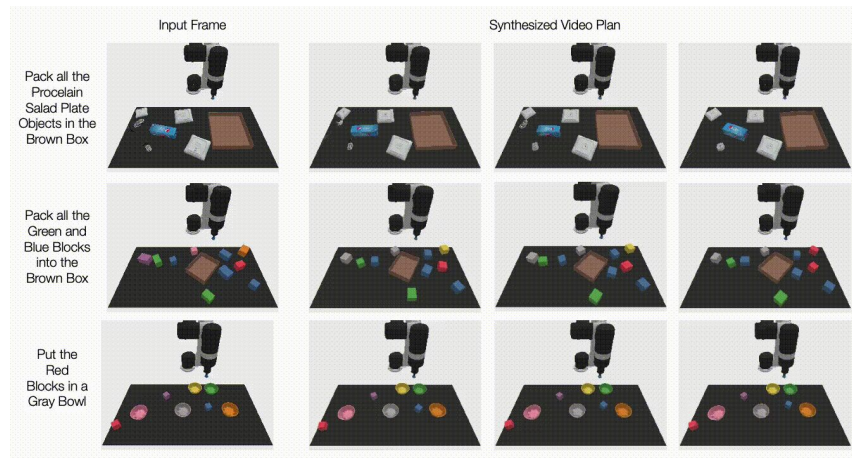


Figure 5

5. Experimental Evaluation

5.3 Real World Transfer

Goal

Evaluate UniPi's ability to **generalize to real-world scenarios**.
Use **large-scale internet video data** to construct **complex action plans**.

Video Synthesis (Quality & Planning Proxy)

- **With pretraining**: significantly better FID/FVD, slightly higher CLIP (Table 4)
- **Without pretraining**: often generates failed plans (Fig. 6)
- **Surrogate success metric**: last-frame classifier
- From scratch: 72.6% / Pretrained+finetuned: 77.1%

Model (24x40)	CLIP Score $\uparrow$	FID $\downarrow$	FVD $\downarrow$	Success $\uparrow$
No Pretrain	24.43 $\pm$ 0.04	17.75 $\pm$ 0.56	288.02 $\pm$ 10.45	72.6%
Pretrain	24.54 $\pm$ 0.03	14.54 $\pm$ 0.57	264.66 $\pm$ 13.64	77.1%

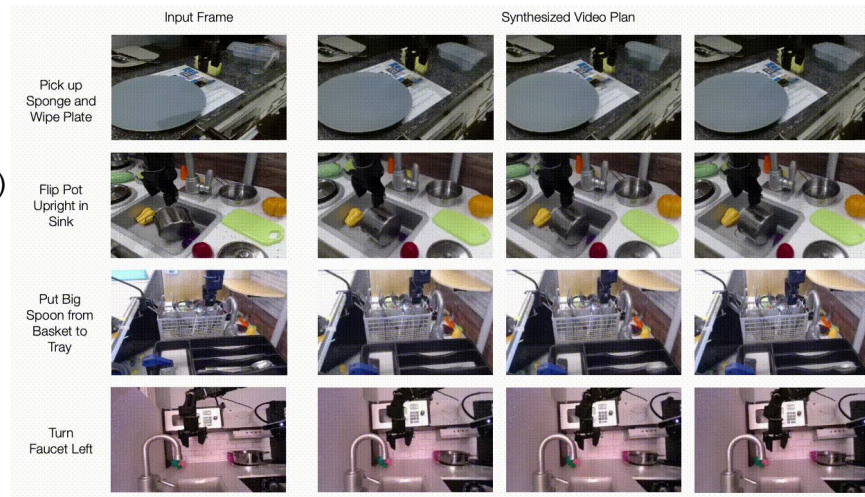
Generalization

Internet pretraining enables UniPi
to handle **novel task commands/scenes** unseen in Bridge (Fig. 8)
Robust to **background changes** (black cropping, photoshopped objects) (Fig. 9)

Key Message

Internet-scale pretraining + robot finetuning

→ higher video quality, better success rates,
stronger generalization & robustness



6. Conclusion

Main Conclusion

Representing policies via **text-conditioned video generation** enables:

Combinatorial generalization

Multi-task learning

Real-world transfer

Suggests: **Generative models + internet-scale data** → foundation for **general-purpose decision-making systems**

Limitations

Slow generation: video diffusion can take minutes for high-quality videos.

- Distillation → **~16× faster** sampling ([44]); further speed-ups possible with efficient samplers.

Fully observed environments: method may hallucinate in **partially observable** settings.

Future Directions

Develop **faster/efficient diffusion** techniques.

Integrate **semantic world knowledge** to reduce hallucination.

Combine **UniPi with LLMs** for richer reasoning and robust planning.